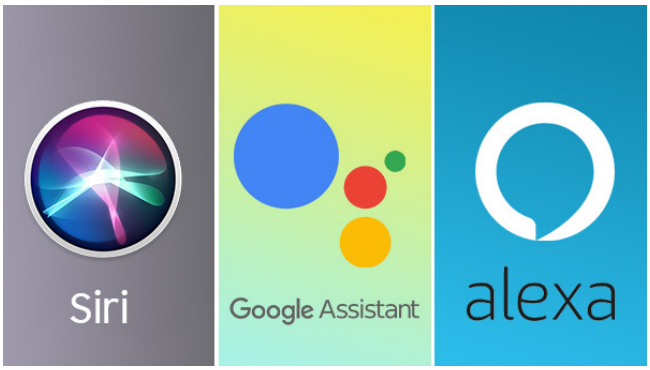
Introduction to Generative AI

What is AI?

- AI, or Artificial Intelligence, refers to machines or computers that can perform tasks that usually require human intelligence.
- These tasks include learning, reasoning, problem-solving, understanding language, and recognizing patterns.
- AI systems are designed to mimic human thinking and decision-making.

Real-Life Examples of AI

1. **Virtual Assistants (Siri, Alexa, Google Assistant)**
    - These AI tools perform tasks like setting reminders, playing music, or answering questions & searching the web.



2. **Recommendation Systems (Netflix, YouTube, Amazon)**
    - Platforms like Netflix, YouTube, and Amazon use AI to suggest movies, videos, or products based on your preferences and past behavior.



3. **Autonomous Vehicles (Self-driving Cars)**
    - Self-driving cars, like those from Tesla, use AI to navigate roads, avoid obstacles, and make driving decisions



4. **Spam Filters (Email)**
    - AI analyzes emails and classifies them as spam or important based on content and user interactions. Detect and filter out unwanted or harmful emails.



What is Generative AI?

- **Generative AI is a type of AI that can create new content, such as images, text, music, or videos.**
- It learns patterns from existing data and generates new outputs based on that learning.

Popular Generative AI Models/LLM.s:

1. Text Generation

These tools specialize in generating human-like text, answering questions, writing essays, or assisting with coding.

- ChatGPT (OpenAI)
- Google Gemini
- DeepSeek
- LLaMA (Meta)
- GitHub Copilot (for code generation)

2. Image Generation

These tools create images, artwork, or designs based on text prompts or other inputs.

- DALL·E (OpenAI)

3. Video Generation

These tools generate or manipulate videos, including animations, and short clips.

Produces realistic videos of people doing or saying things they never actually did.

- Runway ML (for video editing and generation)
- Pictory (creates videos from text)
- DeepBrain (AI video generation)
- DeepFake Lab (for creating deep fake videos)

4. Music Composition

These tools generate original music, beats, or soundtracks based on user inputs or styles.

1. Jukedeck (AI-generated music)

2. OpenAI's MuseNet (generates music in various styles)

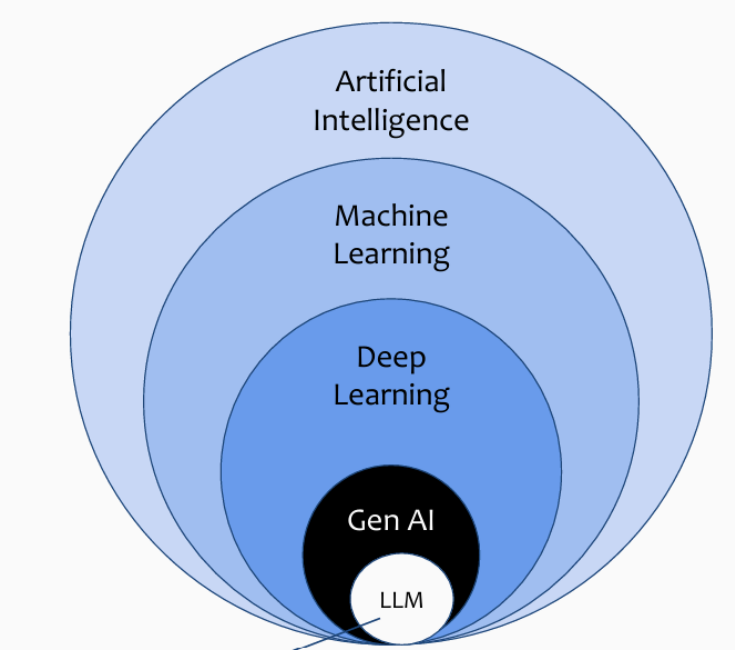
AI Vs Generative AI:

- **AI focuses on performing tasks and making decisions based on data, mimicking human behavior** (e.g., recognizing images, responding to queries).
- **Generative AI specializes in creating or generating new content from learned patterns**, going beyond decision-making to creativity.

LLMs (Large Language Models)

What are LLMs?

- **Large**–as it is trained on large amounts of data and billions of trainable parameters.
- **Language**–as it deals with text data.
- **Model**–as it predicts the next word/sentence/token.



Why QA/Testers Should Learn Generative AI

- **Automated Testing**: Generative AI can create test cases, scripts, and scenarios, saving time and effort.
- **Bug Detection**: AI can help identify potential issues in software by analyzing patterns.
- **Enhanced Creativity**: Generative AI can assist in brainstorming new testing ideas or solutions.
- **Future-Proofing**: As AI becomes more common in software development, learning generative AI ensures QA/testers stay relevant.

Popular Generative AI Models(LLMs)

- **ChatGPT**: A text-based AI model that can answer questions, write content, and assist with coding.
- **Google Gemini**: Google's AI model designed for advanced text generation and problem-solving.
- **DeepSeek**: A generative AI tool focused on creating high-quality text content.
- **GitHub Copilot**: An AI-powered coding assistant that helps developers write code faster and more efficiently.GitHub Copilot.

